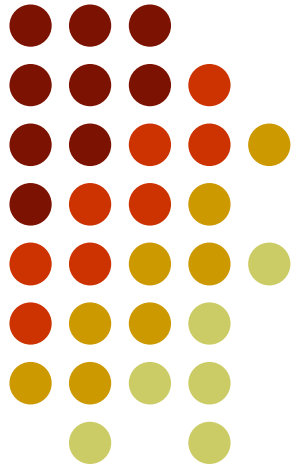
Course Introduction



FINM3013 Introduction to Financial Derivatives

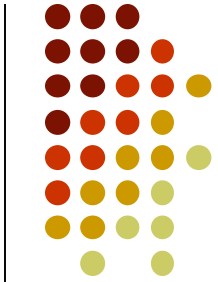


- Course instructor
  - Dr Sherry Zhou; Office: T3-502-R5; Tel.: 3620619
  - Email: sherryzhou@uic.edu.cn
- TA
  - Miss Liang Yue; Office: T3-502-R26; Tel.: 3620743
  - Email: laurenlyue@uic.edu.cn

Office Hours

- Tuesday 16:00-18:00
- Thursday 14:00-18:00
- Friday 10:00-12:00, 14:00-15:00, 17:00-18:00

Objectives



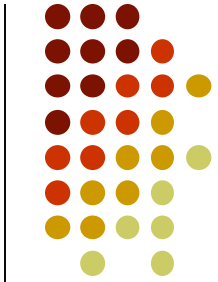
- Give an introduction to students on futures, options and other derivative contracts.
- Focus on the theoretical aspects of these topics.
- Examine the application of the knowledge learnt from the lectures.

Course Content



- Introduction to Derivatives
- Futures
  - Mechanics of Futures Markets
  - Hedging Strategies Using Futures
  - Interest Rate Futures
  - Determination of Future Prices
- Options
  - Mechanics of Options Markets
  - Properties of Stock Options
  - Trading Strategies involving Options
  - Black-Scholes Model

Course Grading



Tasks	Assessment Details	Weighting
Class Participation	To engage students to the coverage of lectures.	**10%**
In-class Exercises	To assess students' understanding of concepts, theories and models.	**30%**
Written Assignments	To measure students' understanding of the theories, models and strategies of derivatives markets.	**20%**
Final Exam	To see how far students have achieved their intended leaning outcomes especially in the knowledge domain.	**40%**

Class Participation

- Factors include
  - Attendance
    - Attendance checking through iSpace
  - Participation in class discussions

In-class Exercises

- Quiz (15%) and Mid-term test (15%)
  - Mid-term test: Chapters 1-6 (Futures)
  - Quizzes
    - Quiz 1: Chapters 1-3; Quiz 2: Chapters 9-11
  - Closed book; Closed notes
  - Formula sheet may be provided
  - Normally no make-up quiz or mid-term test will be given
  - Dates will be informed in advance

Written Assignments

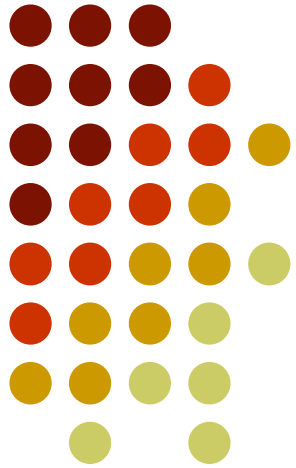
- Group project
  - At most four students per group
  - Topic: TBA
  - No presentation; only report submission through Turnitin
  - **Due date of report: 11:59pm on 20 May 2025**
  - Points will be deducted for late submission (5% deduction for each hour late)
  - Each group only needs one submission of a report. Please list your group members (name and ID) on the front page.

Textbook

- HULL, John C., *Fundamentals of Futures & Options Markets* (8th Edition), Pearson 2013.
- HULL, John C., Options, Futures, and Other Derivatives (9th Edition), Pearson 2018.
- Textbook is not compulsory but you are strongly suggested to have one copy.

Introduction

Chapter 1



The Nature of Derivatives



A derivative is an instrument whose value depends on the values of other more basic underlying variables

Examples of Derivatives



- Futures Contracts
- Forward Contracts
- Swaps
- Options

Why Derivatives Are Important

- Derivatives play a key role in transferring risks in the economy
- There are many underlying assets: stocks, currencies, interest rates, commodities, debt instruments, electricity, insurance payouts, the weather, etc.
- Many financial transactions have embedded derivatives

Futures Contracts



- A futures contract is an agreement to buy or sell an asset at a certain time in the future for a certain price
- By contrast in a spot contract there is an agreement to buy or sell the asset immediately (or within a very short period of time)

Exchanges Trading Futures



- CME Group
- NYSE Euronext
- Eurex
- BM&FBovespa (Sao Paulo, Brazil)
- Tokyo Financial Exchange
- and many more (see list at end of book)

Futures Price



- The futures prices for a particular contract is the price at which you agree to buy or sell at a future time
- It is determined by supply and demand in the same way as a spot price

Electronic Trading



- Traditionally futures contracts have been traded using the open-outcry system where traders physically meet on the floor of the exchange
- This has now been largely replaced by electronic trading and high frequency algorithmic trading is becoming an increasingly important part of the market

Open-Outcry System







Electronic Trading







Examples of Futures Contracts



Agreement to:

- buy 100 oz. of gold @ US$1750/oz. in December
- sell £62,500 @ 1.5500 US$/£ in March
- sell 1,000 bbl. of oil @ US$85/bbl. in April

Terminology



- The party that has agreed to buy has a long position
- The party that has agreed to sell has a short position

Example



- January: an investor enters into a long futures contract to buy 100 oz of gold @ $1,750 per oz in April
- April: the price of gold is $1,825 per oz

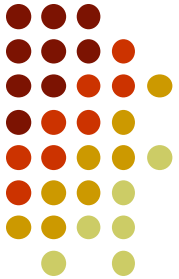
What is the investor's profit or loss?

Over-the-Counter Markets



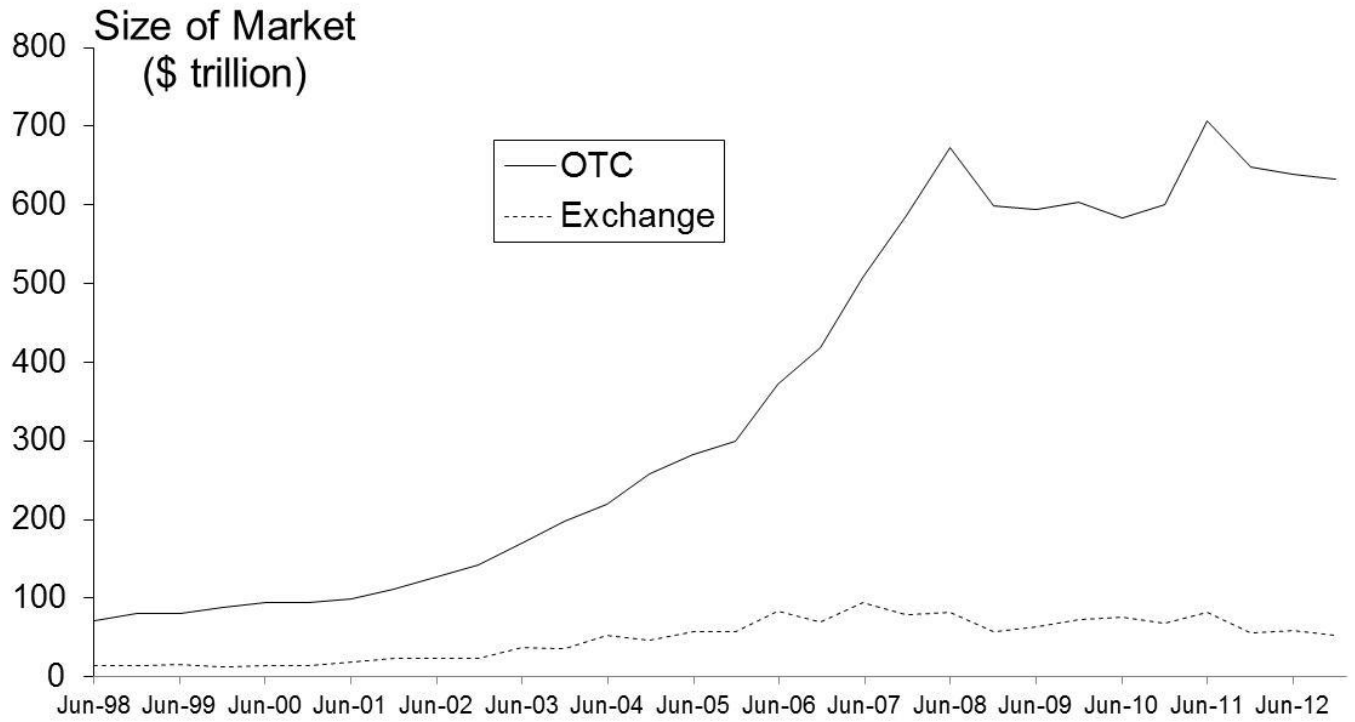
- The over-the counter (OTC) market is an important alternative to exchanges
- Trades are usually between financial institutions, corporate treasurers, and fund managers
- Transactions are much larger than in the exchange-traded market

New Regulations for OTC Market



- The OTC market is becoming more like the exchange-traded market. New regulations introduced since the crisis mean that
  - Standard OTC products in the U.S. must be traded on *swap execution facilities* (SEFs)
  - A central clearing party must be used as an intermediary for standard products
  - Trades must be reported to a central registry

Size of OTC and Exchange-Traded Markets
(Figure 1.2, Page 6)



Source: Bank for International Settlements. Chart shows total principal amounts for OTC market and value of underlying assets for exchange market

Forward Contracts



- Forward contracts are similar to futures except that they trade in the over-the-counter market
- Forward contracts are popular on currencies and interest rates

Forward Price

- The forward price for a contract is the delivery price that would be applicable to the contract if were negotiated today
- The forward price may be different for contracts of different maturities (as shown by the table)

Foreign Exchange Quotes for USD/GBP exchange rate on June 22, 2012 (See Table 1.1, page 7)

	Bid	Offer
Spot	1.5585	1.5589
1-month forward	1.5582	1.5587
3-month forward	1.5579	1.5585
6-month forward	1.5573	1.5580

Example

- On June 22, 2012 the treasurer of a corporation might enter into a short forward contract to sell £100 million in six months at an exchange rate of 1.5573
- This obligates the corporation to pay £100 million and receive $155.73 million on December 22, 2012

Options



- A call option is an option to buy a certain asset by a certain date for a certain price (the strike price)
- A put option is an option to sell a certain asset by a certain date for a certain price (the strike price)

American vs European Options



- An American option can be exercised at any time during its life
- A European option can be exercised only at maturity

Google Call Option Prices (June 25, 2012

Stock Price: bid 561.32, offer 561.51; See page 8)

Strike Price ($)	July Bid	July Offer	Sept Bid	Sept Offer	Dec Bid	Dec Offer
520	46.50	47.20	55.40	56.80	67.70	70.00
540	31.70	32.30	41.60	42.50	55.30	56.20
560	20.00	20.40	30.20	30.70	44.20	45.00
580	11.30	11.60	20.70	21.20	34.50	35.30
600	5.60	5.90	13.50	13.90	26.30	27.10

Google Put Option Prices (June 25, 2012

Stock Price: bid 561.32, offer 561.51; See page 8)

Strike Price ($)	July Bid	July Offer	Sept Bid	Sept Offer	Dec Bid	Dec Offer
520	5.00	5.30	13.60	14.00	25.30	26.10
540	10.20	10.50	19.80	20.30	32.80	33.50
560	18.30	18.70	28.10	28.60	41.50	42.30
580	29.60	30.00	38.40	39.10	51.80	52.60
600	43.80	44.40	51.10	52.10	63.50	64.90

Exchanges Trading Options



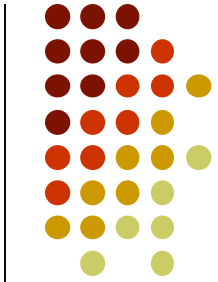
- Chicago Board Options Exchange
- NYSE Euronext
- Boston Options Exchange
- Eurex (Europe)
- and many more (see list at end of book)

Options vs Futures/Forwards



- A futures/forward contract gives the holder the obligation to buy or sell at a certain price
- An option gives the holder the right to buy or sell at a certain price

Three Reasons for Trading Derivatives: Hedging, Speculation, and Arbitrage



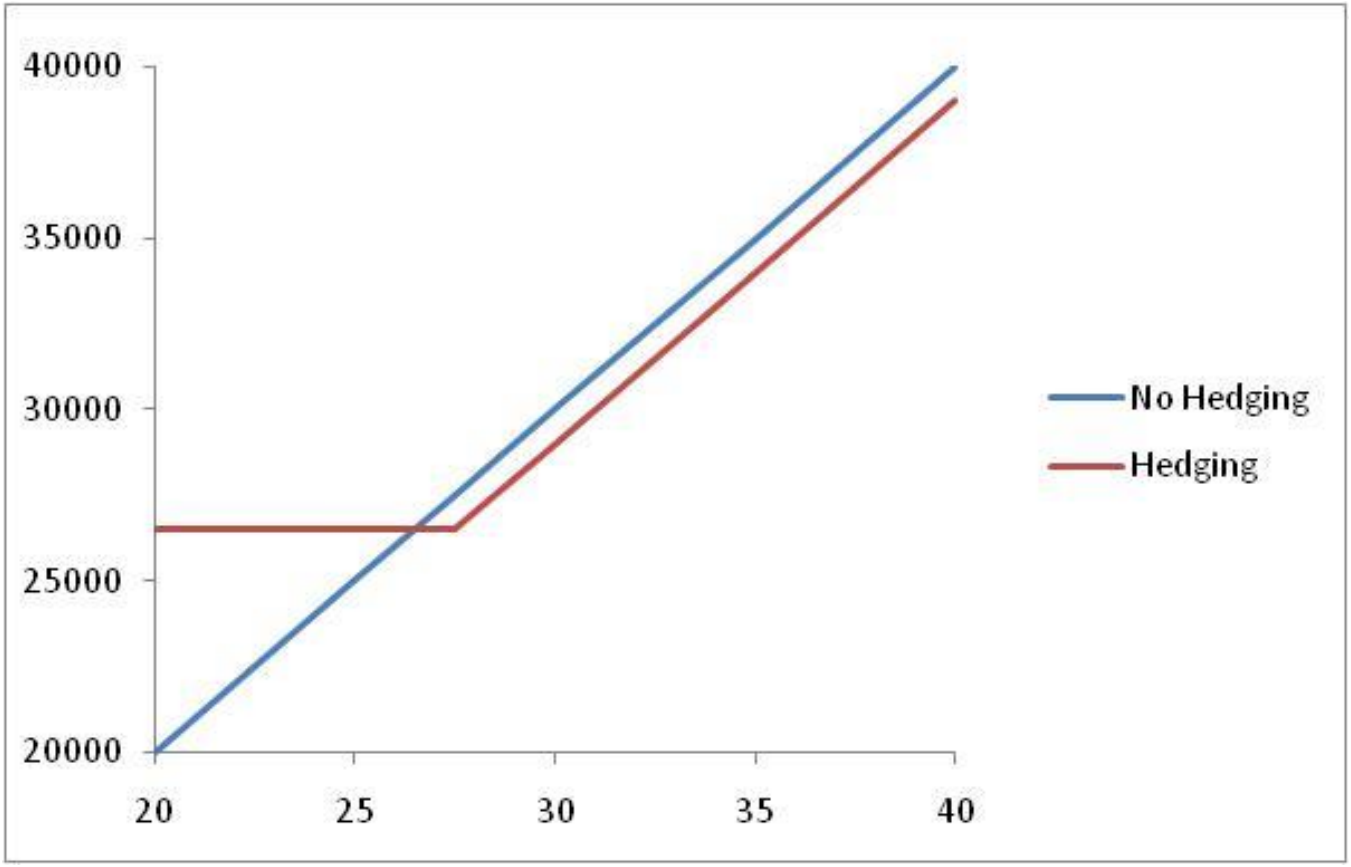
- Hedgers: reduce risks
- Speculators: bet on the future direction of a market variable
- Arbitrageurs: take offsetting positions in two or more instruments to lock in a profit.

Hedging Examples (Example 1.1 and 1.2, page 13)



- A US company will pay £10 million for imports from Britain in 3 months and decides to hedge using a long position in a forward contract
- An investor owns 1,000 shares currently worth $28 per share. A two-month put with a strike price of $27.50 costs $1. The investor decides to hedge by buying 10 contracts

Value of Shares with and without Hedging (Fig 1.4, page 14)



Speculation Example (pages 15)



- A U.S. investor in February thinks that the British pound will strengthen relative to the U.S. dollar over the next two months and is prepared to back that hunch with £250,000.
- An investor with $2,000 to invest feels that a stock price will increase over the next 2 months. The current stock price is $20 and the price of a 2-month call option with a strike of $22.50 is $1.

Speculation Using Futures

Table 1.4 Speculation using spot and futures contracts. One futures contract is on £62,500. Initial margin on four futures contracts = $20,000.

	Possible trades	
	Buy £250,000 *Spot price = 1.4470*	*Buy 4 futures contracts* *Futures price = 1.4410*
Investment	$361,750	$20,000
Profit if April spot = 1.5000	$13,250	$14,750
Profit if April spot = 1.4000	−$11,750	−$10,250

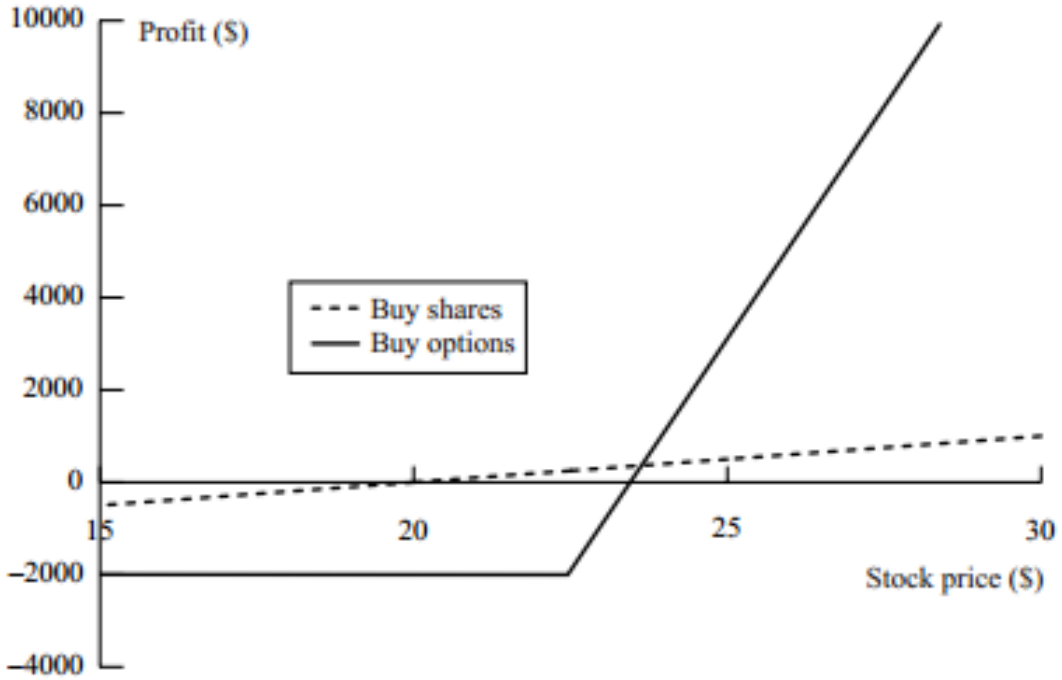
Speculation Using Options

Table 1.5 Comparison of profits from two alternative strategies for using $2,000 to speculate on a stock worth $20 in October.

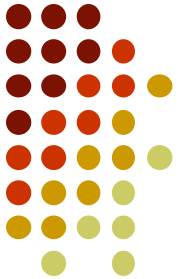
	December stock price	
Investor's strategy	*$15*	*$27*
Buy 100 shares	−$500	$700
Buy 2,000 call options	−$2,000	$7,000

Speculation with Options vs Shares

Figure 1.5 Profit or loss from two alternative strategies for speculating on a stock currently worth $20.



Arbitrage Example (page 17)



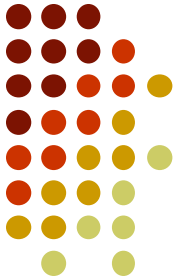
- A stock price is quoted as £100 in London and $152 in New York
- The current exchange rate is 1.5500
- What is the arbitrage opportunity?

Example 1. Gold: An Arbitrage Opportunity?



- Suppose that:
  - The spot price of gold is US$1,700 per ounce
  - The quoted 1-year futures price of gold is US$1,800
  - The 1-year US$ interest rate is 5% per annum
  - No income or storage costs for gold
- Is there an arbitrage opportunity?

The Futures Price of Gold



If the spot price of gold is S & the futures price is for a contract deliverable in T years is F, then

$$F = S(1+r)^T$$

where r is the 1-year (domestic currency) risk-free rate of interest.

In our examples, S=1700, T=1, and r=0.05 so that

$$F = 1700(1+0.05) = 1{,}785$$

Example 2. Oil: An Arbitrage Opportunity?



Suppose that:

- The spot price of oil is US$80
- The quoted 1-year futures price of oil is US$90
- The 1-year US$ interest rate is 5% per annum
- The storage costs of oil are 2% per annum

- Is there an arbitrage opportunity?

Summary

- Futures
  - A futures contract is an agreement to buy or sell an asset at a certain time in the future for a certain price
  - Futures price
  - Long position vs short position
  - Trade in exchanges

Summary



- Forward
  - Forward contracts are similar to futures except that they trade in OTC market
- Options
  - Long call, short call, long put, short put
  - American vs European options
  - Strike price (exercise price)
  - Option premium

Summary

- Options vs Futures/Forwards
  - A futures/forward contract gives the holder the ***obligation*** to buy or sell at a certain price
  - An option gives the holder the ***right*** to buy or sell at a certain price

Summary

- Three types of traders
  - Hedgers: reduce risks
  - Speculators: bet on the future direction of a market variable
  - Arbitrageurs: take offsetting positions in two or more instruments to lock in a profit